

▼ **Figure 45.8 Human endocrine glands and their hormones.** This figure highlights the location and primary functions of the major human endocrine glands. Endocrine tissues and cells are also located in the thymus, heart, liver, stomach, kidneys, and small intestine.

➔ **Mastering Biology Animation:**
Endocrine System Anatomy

Thyroid gland

- **Thyroid hormone (T_3 and T_4):** Stimulates and maintains metabolic processes
- **Calcitonin:** Lowers blood calcium level

Parathyroid glands

- **Parathyroid hormone (PTH):** Raises blood calcium level

Adrenal glands (atop kidneys)

Adrenal medulla

- **Epinephrine and norepinephrine:** Raise blood glucose level; increase metabolic activity; constrict or dilate blood vessels

Adrenal cortex

- **Glucocorticoids:** Raise blood glucose level; control inflammation
- **Mineralocorticoids:** Promote reabsorption of Na^+ and excretion of K^+ in kidneys

Ovaries (in females)

- **Estrogens*:** Stimulate uterine lining growth; promote development and maintenance of female secondary sex characteristics
- **Progesterone*:** Promotes uterine lining growth

Testes (in males)

- **Androgens*:** Support sperm formation; promote development and maintenance of male secondary sex characteristics

*Found in both males and females, but with a major role in one sex

Pineal gland

- **Melatonin:** Participates in regulation of biological rhythms

Hypothalamus

- Hormones released from posterior pituitary (see below)
- **Releasing and inhibiting hormones:** Regulate anterior pituitary

Pituitary gland

Posterior pituitary

- **Oxytocin:** Stimulates contraction of uterus and mammary gland cells; modulates behavior
- **Vasopressin**, also called *antidiuretic hormone (ADH)*: Promotes retention of water by kidneys; modulates behavior

Anterior pituitary

- **Follicle-stimulating hormone (FSH)** and **luteinizing hormone (LH)**: Stimulate gonads (ovaries or testes)
- **Thyroid-stimulating hormone (TSH)**: Stimulates thyroid gland
- **Adrenocorticotropic hormone (ACTH)**: Stimulates adrenal cortex
- **Prolactin:** Stimulates mammary gland growth and milk synthesis in mammals
- **Growth hormone (GH)**: Stimulates growth and metabolic functions
- **Melanocyte-stimulating hormone (MSH)**: Affects color of melanocytes, a type of skin cell

Pancreas

- **Insulin:** Lowers blood glucose level
- **Glucagon:** Raises blood glucose level

such as the thyroid and parathyroid glands and the gonads, either testes in males or ovaries in females (**Figure 45.8**).

Note that endocrine glands secrete hormones directly into the surrounding fluid. In contrast, *exocrine glands* have ducts that carry secreted substances, such as sweat or saliva, onto body surfaces or into body cavities. This distinction is reflected in the glands' names: The Greek *endo* (within) and *exo* (out of) refer to secretion into or out of body fluids, while *crine* (from the Greek word meaning "separate") refers to movement away from the secreting cell. In the case of the pancreas, endocrine and exocrine tissues are found in the same gland: Ductless tissues secrete hormones, whereas tissues with ducts secrete enzymes and bicarbonate.

CONCEPT CHECK 45.1

1. How do response mechanisms in target cells differ for water-soluble and lipid-soluble hormones?
2. What type of gland would you expect to secrete pheromones? Explain.
3. **WHAT IF?** Predict what would happen if you injected a water-soluble hormone into the cytosol of a target cell.

For suggested answers, see Appendix A.

CONCEPT 45.2

Feedback regulation and coordination with the nervous system are common in hormone pathways

Having explored hormone structure, recognition, and response, we now consider how regulatory pathways controlling hormone secretion are organized.

Simple Endocrine Pathways

In a *simple endocrine pathway*, endocrine cells respond directly to an internal or environmental stimulus by secreting a particular hormone. The hormone travels in the bloodstream to target cells, where it interacts with its specific receptors. Signal transduction within target cells brings about a physiological response.